Squatting Biomechanics After Femoroacetabular Impingement Surgery: A Systematic Review and Meta-analysis

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Abstract

Objective: Femoroacetabular impingement syndrome (FAIS) is a prevalent cause of hip pain, characterized by distinctive motion patterns. However, the impact of surgical intervention on squatting biomechanics remains underexplored. This review aims to evaluate the effects of surgical treatment on biomechanical outcomes during squatting in patients with FAIS. **Data sources:** A comprehensive search was conducted across multiple databases following PRISMA guidelines (Prospero registration number CRD42023473974). Data of longitudinal, prospective/retrospective cohorts, case–control, and clinical trial studies with 3-dimensional motion analysis describing kinematics and/or kinetics for any joint during squatting in patients with FAIS comparing presurgical to postsurgical biomechanical data on the same patients were included. Quality assessment was performed using MINORS criteria. Both qualitative and quantitative analyses were performed. **Main results:** Six studies met the inclusion criteria for qualitative review and 5 were suitable for meta-analysis. The overall quality of the studies was moderate based on MINORS criteria. The postsurgical motion analysis was performed between 6 and 32 months postoperatively. Although most studies reported no significant changes in kinematic or kinetic variables postsurgery, individual studies reported specific increases in squat depth, pelvic sagittal range of motion, anterior pelvic tilt, hip and knee flexion, and ankle dorsiflexion angles. Meta-analysis revealed no significant differences in the kinematic parameters analyzed. **Conclusions:** These findings underscore the multifaceted nature of FAIS, suggesting that bone deformities alone do not account for the biomechanical limitations observed during squatting in these patients. This highlights the need for further biomechanical investigations. **Level of evidence:** Systematic Review of Level II–III studies.

Key Words: femoroacetabular impingement, FAIS, motion analysis, squat, kinematic, kinetics

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INTRODUCTION

Femoroacetabular impingement syndrome (FAIS) is a prevalent cause of hip pain, characterized as a motion-related disorder of the hip.[1] It can be categorized as one of the following: (1) Cam-type, consisting of a prominence in the femoral head-neck transition, often seen in athletes, particularly young males in sports like soccer; (2) Pincer-type, consisting of acetabular overcoverage of the femoral head, found frequently in asymptomatic individuals; and (3) Mixed type, a combination of Cam and Pincer deformities, being the most prevalent subtype.[2] The cam-type deformity has been strongly associated with the development of hip osteoarthritis in active, young populations,[3,4] leading to limitations in sports and daily activities and resulting in a decreased quality of life. The economic burden of FAIS can be extensive because it primarily affects the working population, and a recent cost-effectiveness analysis demonstrated a mean cumulative 10-year societal impact of $67,418 per patient treated nonoperatively compared with arthroscopic treatment.[5] Therefore, hip arthroscopy, when performed before the onset of arthritis, can be considered a cost-effective strategy compared with other interventions.[5,6]

In FAIS, a key question revolves around why some people with anatomical deformities never experience symptoms, whereas others with only minor anatomical irregularities suffer from pain and limitations. Motion patterns of the lower limb and trunk may have a role in explaining the variability in FAIS clinical presentations, especially during squatting movements, which bring the proximal femur to the acetabulum rim, leading to impingement. Squatting movements are usually required during everyday activities (eg, getting in and out of a chair or vehicle) and numerous sports; therefore, understanding the biomechanics of squatting in patients with FAIS is crucial.

Individuals with FAIS have been shown to exhibit distinct squatting patterns compared with those without the condition,

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including reduced squatting depth ability and sagittal pelvic range of motion (ROM), greater anterior pelvic tilt, decreased hip internal rotation, and altered hip extensor moments,[7–10] with some conflicting results in the literature.[11] These distinct kinematic and kinetic patterns may represent adaptive mechanisms to reduce pain and stress on the anterior hip or they can be maladaptive features, leading to increased symptoms.

In this context, one critical question arises: can surgical treatment truly modify the movement patterns of FAIS patients? With the advent of arthroscopic surgery, which has demonstrated positive clinical outcomes as a treatment for FAIS,12–14 the rate of surgical interventions for this condition has increased in recent decades, aiming to modify this risk factor and prevent hip arthritis. However, robust evidence regarding the impact of surgery on biomechanical changes is still lacking.

The objective of this review was to evaluate 3-dimensional (3D) in vivo squatting data from the same subjects before and after surgical treatment to investigate whether the biomechanical characteristics of squatting in patients with FAIS may change after surgical treatment compared with presurgical parameters. We hypothesized that surgery would not substantially modify the squat biomechanics in patients postoperatively.

METHODS

Registration

This review was registered with PROSPERO before the commencement of the current study (CRD42023473974).

Eligibility Criteria

Data from longitudinal, prospective/retrospective cohorts, case–control and clinical trial studies with 3D motion analysis describing hip kinematics and/or kinetics during squatting in patients with FAIS comparing presurgical to postsurgical biomechanical data on the same patients were included. No restrictions were imposed on publication date. Exclusions from this systematic review comprised studies lacking 3D motion analysis or not describing kinematics or kinetics. In addition, review articles, case reports, basic science or technical papers, anatomic studies, cadaveric studies or nonhuman investigations, and abstracts were excluded. Studies published in languages other than English, Portuguese, Spanish, French, German, and Italian were excluded. Studies including diagnoses other than FAIS, such as Slipped Capital Femoral Epiphysis sequelae, hip dysplasia, or Legg–Calvé–Perthes disease were also excluded.

Patients diagnosed with FAIS of any age, sex, race, or socioeconomic status and who had been submitted to any type of operative treatment were included in this analysis. Studies analyzing data only of FAIS patients that had been submitted to surgical treatment without baseline presurgical data of the same patients were excluded. Studies comparing FAIS patients with control subjects without a comparison between presurgical and postsurgical biomechanical features were excluded.

Information Sources

Studies were primarily identified through electronic database searches, with secondary identification through expert consultation and scanning reference lists in related articles. A comprehensive literature search was conducted by an experienced medical librarian on December 12, 2024, in the following databases: PubMed/MEDLINE, Embase, Scopus, CINAHL, Google Scholar, and the Cochrane CENTRAL Register of Controlled Trials. The search used both controlled vocabularies (eg, MeSH terms) and keywords regarding the FAI, biomechanics, and surgery. The detailed search strategy is provided in the Supplemental Digital Content 1 (see Material, http://links.lww.com/JSM/A505).

Study Selection

Two investigators (initials blinded for review) independently conducted eligibility assessments, adhering to the reporting guidelines of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA). The Covidence systematic review software (Veritas Health Innovation, Melbourne, Australia; available at www.covidence.org) was used for this process.[15] Initially, titles and abstracts were screened based on inclusion and exclusion criteria. Full-length texts were obtained when titles and abstracts were insufficient for screening and inclusion and exclusion criteria were verified. In addition, the references of all included articles were reviewed to ensure the inclusion of all relevant studies.

Data Collection

An extraction data sheet, containing relevant information, was initially created and subsequently refined after the first round. Two researchers conducted rounds of data extraction, examining tables, graphs, and text. The extracted data from the included studies encompassed authors, year of publication, journal, title, country of study development, population characteristics, study design, study aim, FAI type and diagnosis, surgical technique, inclusion and exclusion criteria, timing from surgery to biomechanical evaluation, data collection and analysis methods, performed tasks, observed primary outcomes, and the main findings and conclusions of the studies. Disagreements were resolved through discussion between the 2 reviewing authors, and if a consensus could not be reached, a senior author made the final decision.

Risk of Bias in Individual Studies

We assessed each study's quality using the Methodological Index for Non-Randomized Studies (MINORS) criteria, a valid instrument designed to assess the methodological quality of nonrandomized surgical studies, whether comparative or noncomparative.[16] Two investigators (initials blinded for review) independently rated each study after full-text analysis, and a mean value was calculated for each of the 12 variables. The score for each variable ranged from 0 to 2, as: 0, if not reported; 1, when reported, but inadequate; and 2, when reported and adequate. Highest total scores indicate lower risk of bias, with 16 the ideal global score for the noncomparative studies and 24 for the comparative studies.[16]

Risk of Bias Across Studies

The risk of bias across studies was assessed by investigating publication bias. Publication date, sample size, and effect size variation in individual studies were reported.

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Summary Measures

Kinematics and kinetics of the hip joint were the primary outcome measures. Other lower limb biomechanical data were also analyzed. For this review, we used internal moments as our standard measure. Therefore, we needed to convert values reported as external moments, so that all the values would be comparable and we could achieve internal coherence.

Methods of Quantitative Analysis

A meta-analysis was performed on 5 of the included studies[9,10,17–19] and 7 outcomes that were reported using similar methodologies. This analysis focused exclusively on studies that reported similar parameters to ensure methodological consistency and to minimize the risk of heterogeneity.

Catelli et al[10,17] and Beaulé et al[20] reported data of the same study population. When data were pooled, only data from Catelli et al[17] were considered for those parameters reported by more than one of these studies. Therefore, data by Catelli et al[10] were used for only 1 of the 7 quantitative outcomes evaluated, and data from Beaulé et al[20] were not used for the meta-analysis. This approach follows the recommendations of the Cochrane Handbook that advice to select the study with the highest quality when similar study populations are reported.[21,22] Catelli et al[17] was considered the study with the highest quality because it received a higher score on the MINORS criteria.

Pooled statistics were computed using inverse variance weighting, assuming a random-effect model and a 95% confidence interval (CI). Mean differences in kinematics between the preoperative and postoperative assessments were the primary measure of treatment effect. Mean differences were not standardized because of the similarity between the units being measured, equipment, and study setting reported across the included studies.

Heterogeneity was quantified using the DerSimonian–Laird method (tau2) and I^2, and χ^2 statistic in accordance with the guidelines delineated in the Cochrane manual.[23] The χ^2 test was interpreted as statistically significant if < 0.05. I^2 statistics followed Cochrane manual for interpretation: 0% to 40% might not be important; 30% to 60% may represent moderate heterogeneity; 50% to 90% may represent substantial heterogeneity; 75% to 100% may represent considerable heterogeneity.[23,24] P-value and 95% CIs were calculated. P-value was considered statistically significant when $P \leq 0.05$. All data analysis was performed using R software (version 4.1.2; R Foundation for Statistical Analysis, Vienna, Austria).

RESULTS

Study Selection

The initial search identified 4612 articles through the databases searched. After removing duplicates, 2149 articles were included for the title and abstract review and 29 articles were selected for full-text review. Finally, 6 studies were included for qualitative review and 5 for quantitative meta-analysis (Figure 1).

Study Characteristics

Methods

The six studies finally selected for the qualitative review had a longitudinal design. Five studies were prospective case–controls and 1 was a randomized controlled trial,[19] and all used 3-dimensional motion analysis with cameras and reflective markers.[10,17–20,25]

Participants

A total of 74 presurgical symptomatic patients with FAIS and 73 postsurgical patients were analyzed. The study population relevant to this review included patients with FAIS treated surgically who had biomechanical data collected both before and after surgery.

The postsurgical motion analysis was taken between 6 and 32 months postoperatively. Two studies analyzed both male and female patients,[18,19] and 4 studies analyzed only male subjects.[10,17,20,25]

Four of these studies had patients submitted to both arthroscopic and open-surgical treatments,[10,17,20,25] and only 2 brought analysis of patients submitted to arthroscopy alone.[18,19] Four of the included studies analyzed patients with cam-type impingement only, and 2 included patients with cam, pincer, or mixed types.[18,19] Detailed data regarding participants are depicted in Table 1.

Outcomes

The outcomes being analyzed were biomechanical characteristics during a squatting task present in patients with femoroacetabular impingement syndrome postsurgical treatment compared with presurgical parameters for the same patients. Details of the surgical procedures used in each study are depicted in Table 1.

Outcomes—Qualitative Synthesis of Results

The characteristics and main findings of each study are depicted in Table 1 and described below. Reported values of kinematic and kinetic parameters described in each study can be found in Tables 2 and 3, respectively.

Kinematics

Qualitative Synthesis

The 6 studies reported kinematic data related to squatting. Beaulé et al,[20] Cvetanovich et al,[18] Grant et al,[19] and Catelli et al[10,17] found no significant differences in maximal squat depth, whereas Lamontagne et al[25] found greater mean maximal squatting depth after surgery.

Beaulé et al,[20] Cvetanovich et al,[18] and Lamontagne et al[25] also found no significant differences for pelvic ROM or any other kinematic parameter. By contrast, Catelli et al[10] reported a significantly greater pelvic sagittal ROM after surgery during both the descent and the ascent phases of the squat. Furthermore, Catelli et al[17] observed a higher hip flexion and an increased anterior pelvic tilt postoperatively during squat descent. These findings differ from Grant et al,[19] who noted reductions in anterior pelvic tilt and hip flexion postoperatively.

In addition, Lamontagne et al[25] analyzed other joints of the lower limbs and reported greater knee flexion, increased ankle dorsiflexion angles, and a larger sum of all joint angles at maximal squat depth. Conversely, Grant et al[19] reported reduced knee flexion postoperatively.

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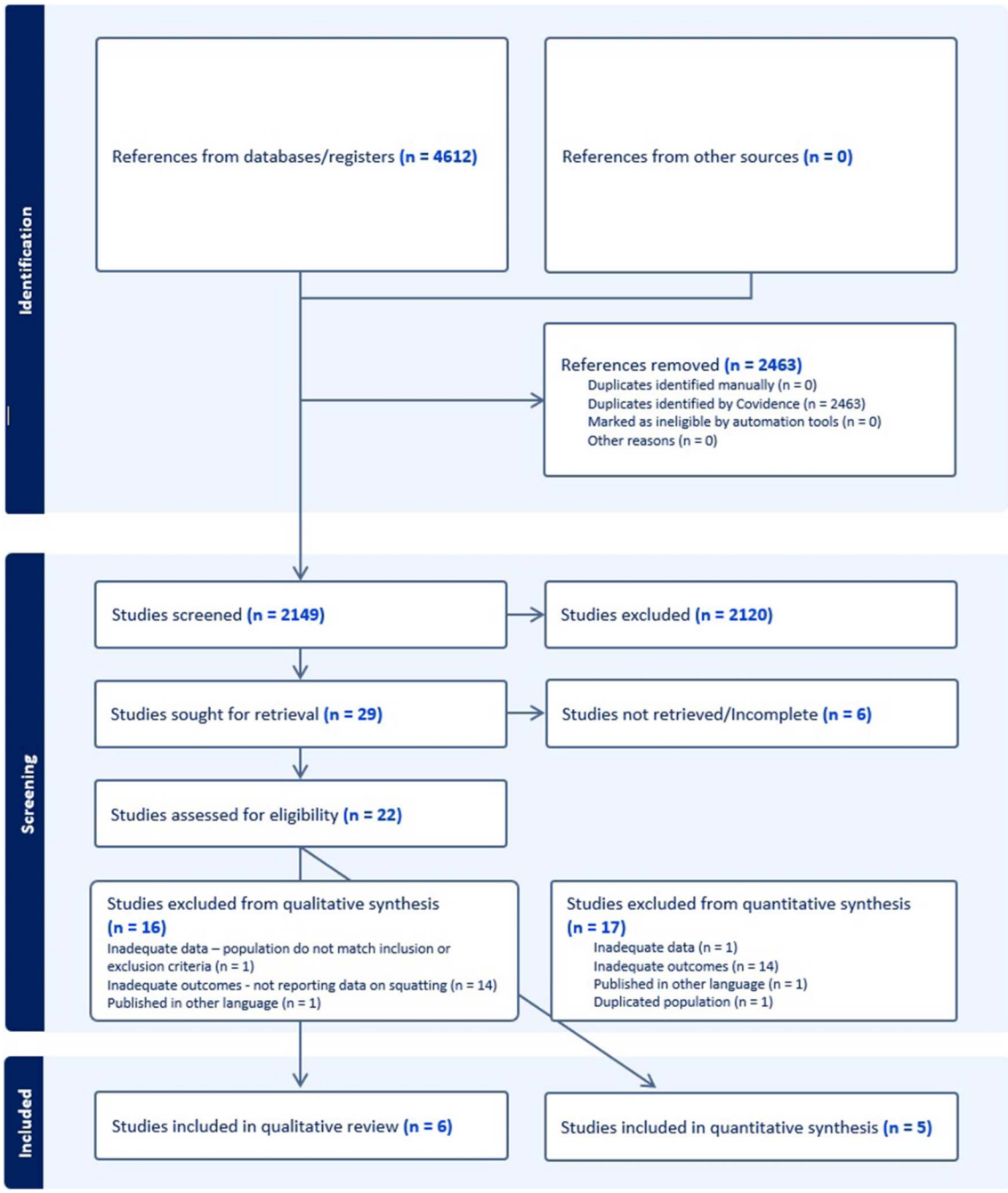

Figure 1. PRISMA 2009 (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) flowchart diagram of the study.

Quantitative Meta-analysis

A meta-analysis was performed by pooling the data from the studies that examined identical kinematic variables during a bilateral squat not repeating the same cohort.[10,17–19,25] These variables included peak hip flexion (Figure 2A), peak knee flexion (Figure 2B), pelvic tilt range of motion (Figure 2C), hip sagittal range of motion (Figure 2D), squat depth (Figure 2E), and squat speed on the descent phase (Figure 2F) and ascent phase (Figure 2G).

Peak Hip Flexion

For peak hip flexion angle, the combined data from the included studies yielded a pooled mean difference of 3.85, with a CI of −4.94 to 12.63 (Figure 2A). The heterogeneity across studies was considerable (I^2 = 75.7%, τ^2 = 58.968, P = 0.006), suggesting inconsistent findings between the studies. The meta-analysis did not find a significant difference between preoperative and postoperative changes in peak hip flexion angle.

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TABLE 1. Studies Characteristics and Main Findings

Author/ Journal/ Year	Study Design	Study Participants	Type of FAIS	Method of Diagnosis	Surgical Technique	Timing of Postoperative Assessment (mo)	Motion Capture	Task	Main Findings Postoperatively vs. preoperatively
Catelli *The Journal of Bone and Joint Surgery* 2020	Prospective case–control	10 male patients with FAIS Age (yr): Presurgery: 35 ± 8 Postsurgery: 37 ± 8 BMI (kg/m^2): Presurgery: 26 ± 3 Postsurgery: 26 ± 4	Cam	Positive impingement test + pelvic and knee CT to confirm the cam morphology (ie, axial [3:00 position] and/or radial [1:30 position] alpha angle of >50.5 degrees and >60 degrees, respectively)	Osteochondroplasty and labral-chondral debridement by: Open dislocation with an anterior approach (n = 3) Combined arthroscopic/mini-open approach (n = 7)	25.2 ± 1.1	10-camera infrared system, 200 Hz (Vicon MX13). 45 retroreflective markers. 2 embedded force plates, 1000 Hz (Bertec Corporation)	Deep double-leg squatting. Five trials at self-selected pace	Full-body kinematics: No significant differences in squatting speed or depth Increased anterior pelvic tilt postoperatively during squat descent (from 11% to 29% of squat cycle (SC), P = 0.004) Higher hip flexion postoperatively than preoperatively (9%-31% SC, P = 0.003) No significant differences for total ROM for: pelvic tilt, hip sagittal, frontal, and transverse planes Hip kinetics: Higher hip extension moment postoperatively compared with the preoperative values during squat descent (16%-17%, P = 0.024 and 21%-26% SC, P = 0.008) No significant differences were reported on the frontal plane Hip contact forces: Higher posterior (77%-81% SC, P = 0.009), superior (67%-71% SC, P = 0.004), and total HCF magnitude (67%-71% SC, P = 0.004) postoperatively in comparison with preoperative values No differences for sagittal and frontal plane HCF vector directions
Lamontagne *The Journal of Bone and Joint Surgery* 2011	Prospective case–control	10 male patients with FAIS Age (yr): Presurgery: 29.9 ± 7.2 Postsurgery: 32.3 ± 7.2 BMI (kg/m^2): Presurgery: 23.3 ± 2.7	Cam	Positive impingement test and a visible cam deformity on anteroposterior and Dunn view radiographs	Open or combined technique involving dislocation of the hip and debridement of the femoral head–neck junction	8-32	9 cameras, 200 Hz (Vicon MX13). 45 retroreflective markers	Deep double-leg squatting. Five trials at self-selected and controlled pace	Kinematics: Greater mean maximal squatting depth postoperatively No significant differences for speed, pelvic angular displacements,

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TABLE 1. Studies Characteristics and Main Findings *(Continued)*

Author/ Journal/ Year	Study Design	Study Participants	Type of FAIS	Method of Diagnosis	Surgical Technique	Timing of Postoperative Assessment (mo)	Motion Capture	Task	Main Findings Postoperatively vs. preoperatively
		Postsurgery: 24.8 ± 3.1							overall pelvic motion, and hip motion Postoperatively: Greater knee flexion, ankle dorsiflexion angles, and sum of all joint angles at maximal squat depth
Cvetanovich *Journal of Hip Preservation Surgery* 2020	Prospective case–control	14 patients with FAIS (13 reevaluated postsurgery) (Females: 11 presurgery, 10 postsurgery) Age (yr): Presurgery: 29.4 ± 9.6 Postsurgery: 29.5 ± 9.5 BMI (kg/m^2): Presurgery: 22.9 ± 4.8 Postsurgery: 23.0 ± 5.2	Cam, pincer, or mixed	Clinical signs and symptoms + AP pelvis radiograph used to measure LCEA and the 90 degrees Dunn lateral view of the hip was used to measure the alpha angle. Preoperative Tonnis grade was also assessed on radiographs	Hip arthroscopy Acetabuloplasty, femoroplasty, and labral repair with T capsulotomy, dynamic fluoroscopic examination, and complete capsular closure by plication	6	12-Optoelectronic camera system, 100 Hz (Qualisys) 28 retroreflective markers 3 floor-embedded force plates, 1200 Hz (Bertec)	Deep - leg squatting and gait Five walking trials for each limb at self-selected speed	Kinematics: No significant differences in any kinematic variable (max. hip flexion, max. hip extension, hip sagittal ROM, descent and ascent velocity) No differences in squat depth Kinetics: Lower postoperative external peak hip extension moment than preoperative No significant differences in any other kinetic variable (hip flexion, adduction, abduction, ER, IR moments, descent and ascent flexion moment rates, and flexion impulse)
Catelli *Journal of Hip Preservation Surgery* 2019	Prospective case–control	11 male patients with FAIS Age (yr): Presurgery: 34.1 ± 7.4 Postsurgery: 36.2 ± 7.4 BMI (kg/m^2): Presurgery: 25.4 ± 2.7 Postsurgery: 25.6 ± 3.6	Cam	Positive impingement test + Radiographs and CT with alpha angles >50.5 degrees and 60 degrees in the oblique axial and radial planes, respectively	Osteochondroplasty of the femoral head–neck junction by: Open approach with surgical dislocation (n = 4) Arthroscopic approach (n = 7)	25.05 ± 1.13	10 infrared cameras, sample rate not informed (Vicon MX13) 45 retroreflective markers Wireless electromyography (EMG) probes, sample rate not informed (BTS FreeEMG 300)	Deep double-leg squatting. Five trials at self-selected and controlled pace	Kinematics: No significant differences in squat depth Postoperative: pelvic sagittal ROM significantly greater for both descent and ascent phases of the squat No significant differences for hip sagittal ROM, peak hip flexion, peak hip abduction, and peak knee flexion
Beaulé *The Journal of Bone and Joint Surgery* 2017	Prospective case–control	10 male patients with FAIS Age (yr): 34.3 (range, 23.1-46.5) BMI (kg/m^2): 26.66 ± 4.79	Cam	Clinical and radiographic examination. Multiplanar CT: Alpha angle of >50.5 degrees at the 3:00 position and/or >60 degrees at the 1:30 position	Chondro-osteoplasty and labral–chondral debridement, with additional labral restabilization in 3 hips and microfracture in 1, by: Hip arthroscopy combined with mini-	24.5 (24–26.6)	10 Vicon cameras, sample rate not informed (MX-13) 45 retroreflective markers 2 Bertec force plates, sample rate not informed (model FP4060-08)	Deep double-leg squatting. Five trials at self-selected and controlled pace	Kinematics: No significant differences in squat depth No significant differences for maximal pelvic tilt ROM and maximal pelvic posterior tilt angle Kinetics:

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TABLE 1. Studies Characteristics and Main Findings *(Continued)*

Author/ Journal/ Year	Study Design	Study Participants	Type of FAIS	Method of Diagnosis	Surgical Technique	Timing of Postoperative Assessment (mo)	Motion Capture	Task	Main Findings Postoperatively vs. preoperatively
					anterior arthrotomy (n = 6) Open surgical hip dislocation (n = 3) Hip arthroscopy (n = 1)				No significant differences for max. hip flexion moment, max. hip extension moment, max. hip power generation, and max. hip vertical force
Grant *PeerJ* 2024	Randomized controlled trial	19 patients with FAIS (females: 9) Age (yr): Presurgery: 33.9 ± 11.7 Postsurgery: 34.9 ± 11.8 BMI (kg/m^2): Presurgery: 23.4 ± 3.3 Postsurgery: 23.7 ± 3.2	Cam, pincer, or mixed	Clinical signs + radiographs. Cam morphology (alpha angle >55 degrees) and/or pincer morphology (positive crossover sign and/or lateral center-edge angle >40 degrees) in imaging	Hip arthroscopy; bone reshaping; labral and cartilage pathologies were treated	12.1 ± 1.3	12-camera motion capture system, 120 Hz (Vicon) Reflective markers 2 ground-embedded force plates, 1200 Hz (AMTI)	Deep double-leg squatting Five trials at self-selected pace and in standardized footwear	Kinematics: No significant differences in squat depth -Slower squat speed postoperatively Greater trunk flexion postoperatively Reduced anterior pelvic tilt, hip flexion, and knee flexion postoperatively Kinetics: Lower postoperative external hip flexion moments during squat descent and ascent

AP, anteroposterior; BMI, body mass index; CT, computed tomography; FAIS, femoroacetabular impingement syndrome; HCF, hip contact forces; LCEA, lateral center edge angle; ROM, range of motion; SC, squat cycle.

Peak Knee Flexion

For peak knee flexion angle, the pooled mean difference was 0.25, with a 95% CI of −6.49 to 6.99 (Figure 2B). This suggests that there is no significant difference between preoperative and postoperative conditions overall. The heterogeneity of these studies was low (I^2 = 0%, τ^2 = 0, P = 0.369), indicating a low level of variability in outcomes between the studies.

Pelvic Tilt Range of Motion

For pelvic tilt range of motion, the pooled mean difference was −0.06, with a 95% CI of −3.38 to 3.27 (Figure 2C). There was no significant difference between preoperative and postoperative conditions. The heterogeneity among the studies was low (I^2 = 0%, τ^2 = 0.00, P = 0.728), suggesting that the effect sizes are consistent across the studies.

Hip Sagittal Range of Motion

For hip sagittal range of motion, the pooled mean difference was 3.13, with a 95% CI of −3.81 to 10.07 (Figure 2D). There was no significant difference between preoperative and postoperative conditions. The heterogeneity among the studies was low (I^2 = 0%, τ^2 = 0.00, P = 0.407), suggesting that the effect sizes are consistent across the studies.

Squat Depth

For squat depth, the pooled mean difference was 1.46, with a CI of −3.43 to 6.35 (Figure 2E). There was no significant difference when comparing preoperative to postoperative values overall. The heterogeneity between the studies was low (I^2 = 0%, τ^2 = 0.00 P = 0.819), suggesting similar variability in the results of the included studies.

Squat Speed—Descent Phase

For squat speed on the descent phase, the pooled mean difference was 0.02 with a 95% CI of −0.05 to 0.08 (Figure 2F). This suggests no significant difference between preoperative and postoperative values overall. Heterogeneity may represent moderate heterogeneity (I^2 = 40.2%, τ^2 = 0.001, P = 0.188), but with a P-value of 0.188, that is not statistically significant. This indicates some variability in the effects measured by the studies that was not solely because of chance.

Squat Speed—Ascent Phase

For squat speed on the ascent phase, the pooled mean difference was 0.0, with a 95% CI from −0.07 to 0.08 (Figure 2G). This implies that there was no significant difference between the preoperative and postoperative values across the studies. The heterogeneity among the studies was moderate (I^2 = 48.5%, τ^2 = 0.002, P = 0.143), indicating variability in effect sizes across the studies.

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TABLE 2. Numerical Values of Kinematic Parameters Described in Each Study

	Pre	SD	Post	SD	Subjects	Phase of Squat	*P*
Squat depth (% of leg length)							
Catelli 2020	30.5	8.9	31.3	13.4	10	Maximal depth	>0.05
Lamontagne 2011	36.9	12	33.2	10.3	10	Maximal depth	0.027
Catelli 2019*	33.1	11.4	34.2	15.6	11	Maximal depth	0.05
Beaulé 2017*	44	6.8	37	9.9	10	Maximal depth	>0.05
Cvetanovich 2020†	0.34	0.10	0.36	0.10	15	Maximal depth	>0.05
Grant 2024	43.11	10.27	41.67	11.24	19	Maximal depth	>0.05
Squat speed (m/s)—descent							
Catelli 2020	0.27	0.14	0.24	0.06	10	Descent	>0.05
Cvetanovich 2020	0.4	0.2	0.5	0.2	15	Descent	0.080
Grant 2024	0.24	0.08	0.2	0.06	19	Descent	<0.05
Squat speed (m/s)—ascent							
Catelli 2020	0.30	0.14	0.30	0.08	10	Ascent	>0.05
Cvetanovich 2020	0.4	0.2	0.5	0.2	15	Ascent	0.100
Grant 2024	0.3	0.1	0.25	0.08	19	Ascent	<0.05
Trunk sagittal ROM							
Grant 2024	22.1	12.9	27	11.6	19	Entire cycle	>0.05
Pelvic tilt (ROM)							
Catelli 2019	9	4.5	16	6.2	11	Descent	0.016
Catelli 2019	8.9	3.4	14.7	7.3	11	Ascent	0.047
Catelli 2020	24	7	26	7	10	Entire cycle	0.550
Lamontagne 2011*	17.6	12.4	18.6	15.1	10	Entire cycle	>0.05
Grant 2024	16.17	6.89	15.21	6.25	19	Entire cycle	>0.05
Peak posterior pelvic tilt							
Beaulé 2017	11.6	6.65	13.9	5.45	10	Entire cycle	0.782
Peak hip flexion							
Catelli 2020‡	117	7.5	120.3	6.1	10	Entire cycle PEAK	
Lamontagne 2011	109.8	9.7	106.2	8.8	10	At maximal squat depth PEAK	>0.05
Cvetanovich 2020	98.5	19.6	99.2	16.7	15	Entire cycle PEAK	0.950
Catelli 2019	95.4	19.5	104.1	8.8	11	Entire cycle PEAK	0.054
Grant 2024*	80.98	13.96	64.79	15.08	19	Entire cycle PEAK	<0.05
Peak hip extension (degrees)							
Cvetanovich 2020	3.9	5	2.7	3.5	15	Entire cycle	0.060
Hip sagittal ROM							
Catelli 2020	113	12	113	13	10	Entire cycle	0.980
Cvetanovich 2020	94.7	22.9	96.6	17.4	14 pre, 13 post	Entire cycle	0.610
Catelli 2019	91.4	24.2	101.3	7.2	11	Entire cycle	>0.05
Grant 2024	80.43	17.15	71.46	17.69	19	Entire cycle	>0.05
Peak hip abduction (degrees)							
Catelli 2019	13.3	6.2	11.6	4.8	11	Entire cycle	>0.05
Hip frontal ROM							
Grant 2024	9.22	6.08	10.29	4.14	19	Entire cycle	>0.05
Hip transverse ROM							
Grant 2024	15.78	6.39	18.73	9.68	19	Entire cycle	>0.05
Peak knee flexion (degrees)							
Catelli 2019	123.0	15.1	121.3	20.3	11	Maximal depth	>0.05
Lamontagne 2011	130.6	19.4	141.5	19.2	10	Maximal depth	0.005
Grant 2024*	109	13.31	106.44	13.23	19	Maximal depth	>0.05
Knee sagittal ROM							

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TABLE 2. Numerical Values of Kinematic Parameters Described in Each Study *(Continued)*

	Pre	SD	Post	SD	Subjects	Phase of Squat	*P*
Grant 2024	107.4	15.4	105.55	12.51	19	Entire cycle	>0.05
Ankle dorsiflexion (degrees)							
Lamontagne 2011	22.8	4	30.2	7.5	10	Maximal depth	0.048
Ankle sagittal ROM							
Grant 2024	43.06	7.02	44.3	6.56	19	Entire cycle	>0.05

** Mean values and/or SD calculated from graphs.*
† Data in meters informed by the author.
‡ Data sent by author.
ROM, range of motion.

Kinetics

Four articles reported data on squatting kinetics.[17–20] Although Cvetanovich et al[18] reported a lower postoperative external peak hip extension moment than preoperative values (Pre: 0.4 ± 0.5 vs Post: 0.1 ± 0.4 N·m/BW·H; $P = 0.006$),[18] Catelli et al[17] reported a higher hip extension moment postoperatively compared with the preoperative values during squat descent (16% to 17%, $P = 0.024$ and 21% to 26% SC, $P = 0.008$),[17] and Grant et al[19] reported lower external hip flexion moments at follow-up (descent: −0.84 N·m/BW·HT [%]; ascent: −0.90 N·m/BW·HT [%]). Regarding other kinematic parameters, no significant differences were observed for hip joint moments in the sagittal, frontal, and transverse planes.[17–20]

Hip Contact Forces

One study reported data on hip contact forces (HCFs) during squatting.[17] They found higher posterior (77% to 81% SC, $P = 0.009$), superior (67% to 71% SC, $P = 0.004$), and total

TABLE 3. Numerical Values of Kinetic Parameters Described in Each Study (Values Described as Internal Moments)

	Pre	SD	Post	SD	Subjects	Phase of Squat	Units	*P*
Hip flexion moment								
Catelli 2020*	−0.048	0.090	−0.042	0.075	10	Entire cycle	Nm/kg	
Cvetanovich 2020	0.400	0.500	0.100	0.400	14 pre; 13 post	Entire cycle	N·m/BW·h	0.006
Beaulé 2017	−0.020	0.090	0.040	0.110	10	Entire cycle - maximal moment	Nm/kg	0.279
Hip extension moment								
Catelli 2020*	−0.574	0.138	−0.625	0.135	10	Entire cycle	Nm/kg	
Cvetanovich 2020	−6.400	1.600	−6.300	1.700	14 pre; 13 post	Entire cycle	N·m/BW·h	0.770
Beaulé 2017	−0.720	0.070	−0.790	0.280	10	Entire cycle—maximal moment	Nm/kg	0.459
Hip abduction moment								
Cvetanovich 2020	−2.400	1.600	−1.900	2.000	14 pre; 13 post	Entire cycle	N·m/BW·h	0.060
Hip adduction moment								
Cvetanovich 2020	1.200	0.900	1.100	1.100	14 pre; 13 post	Entire cycle	N·m/BW·h	0.710
Hip external rotation moment								
Cvetanovich 2020	−0.800	0.800	−0.500	0.500	14 pre; 13 post	Entire cycle	N·m/BW·h	0.180
Hip internal rotation moment								
Cvetanovich 2020	0.500	0.500	0.400	0.700	14 pre; 13 post	Entire cycle	N·m/BW·h	0.670
Max hip power generation								
Beaulé 2017	0.87	0.18	1.08	0.45	10	Entire cycle—maximal	W/kg	0.279
Max hip vertical force								
Beaulé 2017	3.50	0.36	3.93	0.97	10	Entire cycle—maximal	N/kg	0.155
Extension impulse								
Cvetanovich 2020	−12.3	4.5	−9.6	3.5	14 pre; 13 post	Entire cycle	N·m/BW·h	0.100
Descent extension moment rate								
Cvetanovich 2020	−2.7	1.5	−3.6	1.5	14 pre; 13 post	Entire cycle	N·m/BW·h	0.100
Ascent extension moment rate								
Cvetanovich 2020	−3.5	1.6	−3.9	1.9	14 pre; 13 post	Entire cycle	N·m/BW·h	0.540

** Data sent by the author in N·mm/BW and transformed to N·m/kg.*

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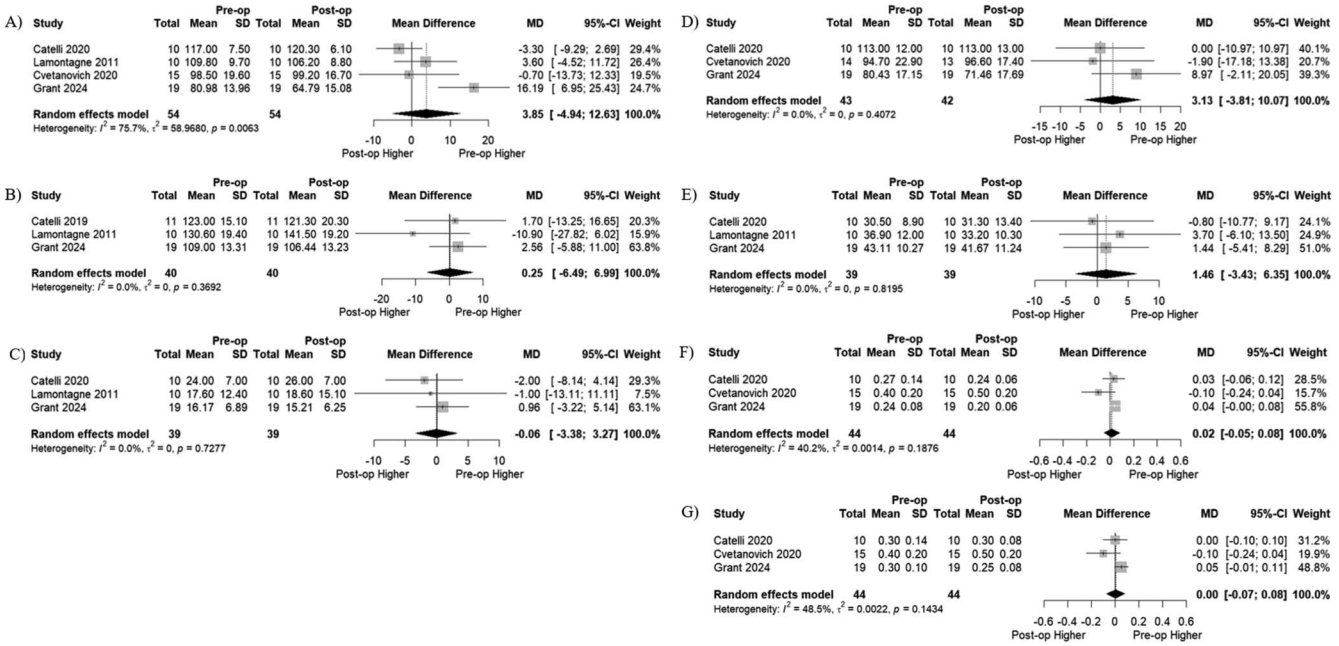

Figure 2. Standardized mean differences between FAIS patients preoperatively and postoperatively for (A) peak hip flexion; (B) peak knee flexion; (C) pelvic tilt ROM; (D) hip sagittal ROM; (E) squat depth; (F) squat speed on the descent phase; and (G) ascent phase.

HCF magnitude (67% to 71% SC, $P = 0.004$) postoperatively in comparison with preoperative values.

Risk of Bias Within Studies

Considering 16 the ideal global score for noncomparative studies and 24 for comparative studies, the overall quality of the studies included in this systematic review was moderate based on MINORS criteria (Figure 3).

Risk of Bias Across Studies

Evidence of publication bias among the included studies remains inconclusive. Because of the limited number of studies, funnel plots were deemed inappropriate for assessing publication bias and, therefore, were not included in this analysis. Publication date, sample size, and effect size variation in individual studies were reported in Table 1.

DISCUSSION

This systematic review and meta-analysis sheds light on the biomechanical impacts of FAIS surgery on squatting, revealing intriguing findings regarding kinematic and kinetic parameters. This study provides preliminary evidence suggesting that FAIS surgery may not substantially modify the squat biomechanics in patients postoperatively. Although the meta-analysis found no significant changes in kinematic variables after surgery, the qualitative assessment of kinematic and kinetic variables revealed inconsistent results across studies. These findings highlight that kinetic and kinematic alterations present before surgery could persist afterward, which might help to explain the variability in clinical outcomes after surgery.

This review uncovered several noteworthy observations regarding kinematic parameters. Existing literature documents reduced squat depth in individuals with cam-type impingement[7–9] but, despite the surgical correction of bone deformities, the majority of studies did not observe significant improvements in squat depth.[10,17–20] This suggests that the depth of squatting among individuals with hip impingement may not solely stem from bone deformities. Nevertheless, it is noticeable that patients in the reviewed studies achieved squatting depths comparable to those of controls reported in the literature,[8,9,26] indicating a potentially limited scope for postsurgical improvement.

However, a notable exception was observed in the study by Lamontagne et al,[25] which reported improvements in squatting performance postsurgery, attributed to increased knee flexion, ankle dorsiflexion, and a more posterior pelvic tilt during the descent phase. These results suggest potential biomechanical improvements facilitated by surgery, possibly linked to decreased joint pain. Although Grant et al[19] also observed a reduced anterior pelvic tilt throughout the descent and ascent phases postoperatively,[19] Catelli et al[17] described an increased anterior pelvic tilt throughout the squat cycle after surgery, resembling data from the healthy group of this study, which was suggested to possibly be because of the removal of bone deformities facilitating a return to a more natural anterior pelvic tilt in these patients.

Despite observed limitations in pelvic sagittal ROM during squatting,[9] only one study highlighted greater pelvic sagittal ROM postsurgery.[10] Surprisingly, Grant et al[19] found reduced hip flexion throughout the squat cycle postoperatively. This reduction may reflect a learned mechanism to avoid impingement, corresponding more likely to the natural course of FAIS rather than a direct outcome of surgical treatment, as Grant et al[19] suggested. In addition, all other pelvic and hip kinematic variables analyzed did not show significant changes after surgery, emphasizing the variability in outcomes across different kinematic parameters.

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Regarding kinetic parameters, this review identified intriguing changes postsurgery. Cvetanovich et al[18] reported a reduced postoperative external peak hip extension moment compared to preoperative values, potentially indicating reduced demand on the hip flexors, which could contribute to pain reduction. Conversely, Catelli et al[17] observed greater hip extension moments postsurgery during squat descent, and Grant et al[19] identified lower postoperative external hip flexion moments during squat ascent and descent, although these changes were no longer evident after correcting for squatting speed. These alterations may suggest a correction of muscle force imbalances present preoperatively.

Despite these findings, no other significant differences were observed for hip joint moments in sagittal, frontal, and transverse planes across multiple studies,[17,18,20] highlighting the complexity of the biomechanical changes associated with FAI surgery.

Finally, it is important to note that five studies evaluated patient-reported outcomes after surgery, consistently demonstrating improvements in these scores, from 6 months to 2 years of follow-up, among patients who underwent surgical treatment. Grant et al,[19] for instance, reported significant postoperative improvements in the International Hip Outcome Tool questionnaire (iHOT-33) and subscales of the Hip Disability and Osteoarthritis Outcome Score (HOOS), including pain, symptoms, activities of daily living (ADL), sports and recreational activities (SRA), and quality of life (QoL) at the 12-month follow-up. Similarly, Catelli et al[10,17] observed improvements at 2 years of follow-up across all parameters evaluated by the HOOS questionnaire. Beaulé et al[20] highlighted significant gains in overall HOOS symptom scores at the 2-year follow-up, particularly in pain, ADL, SRA, and QoL subscales. In addition, Cvetanovich et al[18] reported enhanced hip-specific functional outcomes, including the Hip Outcome Score Sport-Specific Subscale (HOS-SS) and the Activities of Daily Living Subscale (HOS-ADL) at the 6-month follow-up. Although improvements in patient-reported outcomes after surgery were consistently observed, it is possible that the positive results were influenced not only by the surgical intervention but also by postoperative rehabilitation efforts focusing on addressing muscle imbalances.[10,17,18] Therefore, a more comprehensive analysis of both treatment modalities—surgery and rehabilitation—would provide valuable insights for clinicians aiming to optimize patient care and outcomes.

									COMPARATIVE STUDIES				
	A CLEARY STATED AIM	INCLUSION OF CONSECUTIVE PATIENTS	PROSPECTIVE COLLECTION OF DATA	ENDPOINT APPROPRIATE TO STUDY AIM	UNBIASED ASSESMENT OF THE STUDY ENDPOINTS	FOLLOW-UP PERIOD APPROPRIATE	LOSS TO FOLLOW UP LESS THAN 5%	PROSPECTIVE CALCULATION OF THE STUDY SIZE	AN ADEQUATE CONTROL GORUP	CONTEMPORARY GROUPS	BASELINE EQUIVALENCE OF GROUPS	ADEQUATE STATISCAL ANALYSES	**TOTAL**
Catelli et al. The Journal of Bone and Joint Surgery 2020	2	1	2	2	0	2	1.5	0	2	2	2	2	**18.5**
Lamontagne et al. The Journal of Bone and Joint Surgery 2011	2	1	1.5	2	0	1	1	0	2	1.5	2	1.5	**15.5**
Cvetanovich et al. Journal of Hip Preservation Surgery 2020	2	2	2	2	0	1.5	2	0	2	1	2	2	**18.5**
Catelli et al. Journal of Hip Preservation Surgery 2019	2	1	2	2	0	2	1	0	2	2	2	2	**18**
Beaulé et al. The Journal of Bone and Joint Surgery 2017	2	1	2	2	0	2	1	0	N/A	N/A	N/A	N/A	**10**
Grant et al. PeerJ 2024	2	1	2	2	1	2	1.5	0	2	2	2	2	**19.5**

Figure 3. Risk of bias based on MINORS criteria.

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The findings from our article suggest that bone deformities are not the sole cause of altered squatting biomechanics in FAIS patients, which might also be linked to other factors such as pain and altered muscle balance and neuromuscular control. In addition, the heterogeneity of biomechanical patterns observed across studies might suggest that FAIS patients do not behave as a homogeneous group, and identifying different biomechanical subgroups is an important step to understand the diverse adaptations these patients may adopt to the pathology, and how these biomechanical differences could obscure the effects of surgical interventions in specific subgroups of patients. Future studies should investigate the presence of biomechanical subgroups and if the persistence of movement dysfunctions linked to FAIS mechanical etiopathogenesis may be a risk factor for disease progression and worse clinical outcomes, and if physical therapy interventions can improve these movement dysfunctions. Investigating how surgery affects other daily life tasks, such as gait, is also crucial for understanding how FAIS surgery can impact different outcomes.

Limitations

Some limitations present in the included studies and in this review require acknowledgement. A relatively few studies and patients were included because of the limited publications in this field, and the representation of women was limited. The usage of the similar cohort of participants by 3 of the included studies[10,17,20] limited the power of our conclusions. Some inconsistencies regarding data reporting of these studies were also observed. In addition, methods for data processing of kinematics were different between articles, which might have some influence over the numerical results. Almost all the included studies were case–control, cross-sectional studies of moderate quality. Moreover, most participants included were recruited from orthopedic clinics, which may reflect more severe presentations of the condition. Finally, the lack of standardization in surgical techniques across the studies should be noted. Adjusting the analysis to account for uniform surgical methods could potentially yield different outcomes.

CONCLUSIONS

This systematic review and meta-analysis provide valuable insights into the biomechanical effects of FAIS surgery on squatting. Although the meta-analysis revealed no significant differences in the kinematic parameters analyzed, the qualitative review showed some variability across different studies, hence providing a complex overview. These findings underscore the multifaceted nature of FAIS, suggesting that bone deformities alone do not account for the biomechanical limitations observed during squatting in these patients. Future investigations should further clarify the clinical implications of these findings and explore additional factors influencing biomechanical outcomes after FAIS surgery.

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